



DevOps Theory CA2

Causal Adaptive Multi Agent Transport Intelligent System

DevOps Architecture & Deployment Reflection Report

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Microservice Architecture Built on FastAPI, Docker & Kubernetes

System Architecture Highlights:

- Ansible automates host provisioning & environment setup.
- FastAPI backend containerized using Docker (Python 3.11-slim).
- Kubernetes manages 2 backend replicas behind a K8s Service (Port 8000).

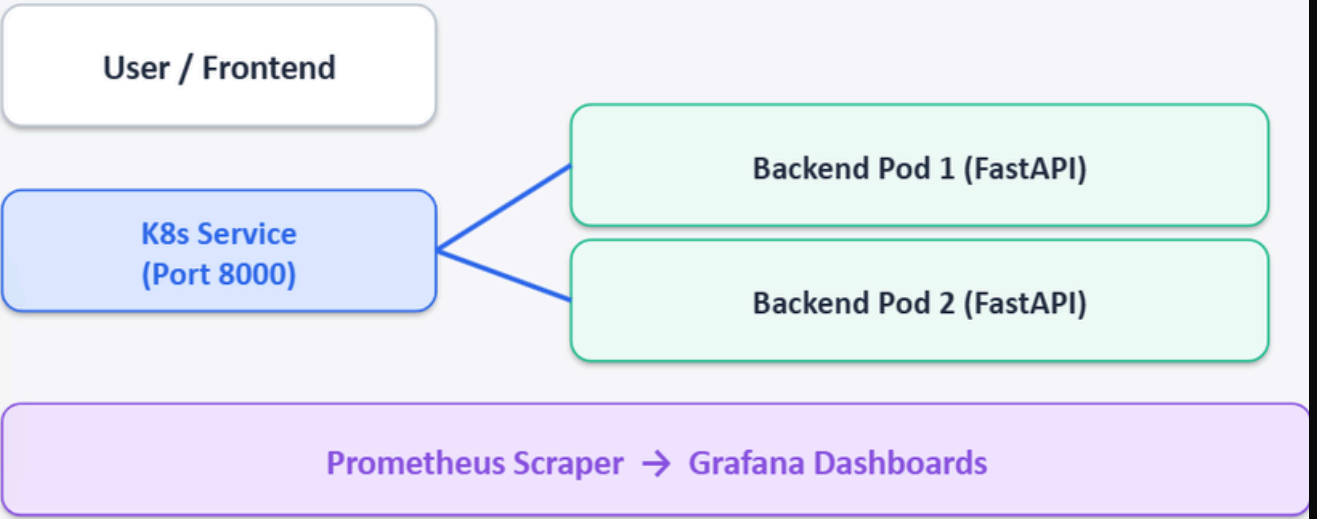
Kubernetes Cluster Deployment Status

```
PS C:\Users\Sanskriti\OneDrive\Desktop\CAMATIS Devops> kubectl get pods,deploy,svc -n camatis -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP              NODE                   NOMINATED NODE   READINESS GATES
pod/camatis-backend-56584c9948-qzqqk 1/1     Running   0           40m   10.244.0.9      desktop-control-plane   <none>           <none>
pod/camatis-backend-56584c9948-wfm9p 1/1     Running   0           39m   10.244.0.10     desktop-control-plane   <none>           <none>
pod/grafana-57b7966f5-gdt25           1/1     Running   0           5m30s 10.244.0.13     desktop-control-plane   <none>           <none>
pod/prometheus-76f96f4ffd-q77rv       1/1     Running   0           12m   10.244.0.12     desktop-control-plane   <none>           <none>

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE   CONTAINERS       IMAGES                                     SELECTOR
deployment.apps/camatis-backend      2/2     2             2           124m   camatis-backend  ghcr.io/sansshr/camatis-backend:latest  app=camatis-backend
deployment.apps/grafana              1/1     1             1           5m30s  grafana          grafana/grafana:11.4.0                  app=grafana
deployment.apps/prometheus           1/1     1             1           29m   prometheus       prom/prometheus:v2.55.1                  app=prometheus

NAME                                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE   SELECTOR
service/camatis-backend             ClusterIP    10.96.145.19 <none>        8000/TCP   124m   app=camatis-backend
service/grafana                     ClusterIP    10.96.253.4  <none>        3000/TCP   5m30s  app=grafana
service/prometheus                  ClusterIP    10.96.210.251 <none>        9090/TCP   29m    app=prometheus
PS C:\Users\Sanskriti\OneDrive\Desktop\CAMATIS Devops> |
```

CAMATIS — Kubernetes Architecture & Observability



Ansible Idempotency Verification

```
TASK [VERIFY | Show camatis user info] *****
ok: [localhost]

TASK [VERIFY | Display user info] *****
ok: [localhost] => {
  "msg": "uid=1001(camatis) gid=1001(camatis) groups=1001(camatis)"
}

PLAY RECAP *****
localhost : ok=11  changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

sanskriti11@LAPTOP-LS41B23P:/mnt/c/Users/Sanskriti/OneDrive/Desktop/CAMATIS Devops/ansible$ |
```

GitHub Actions Pipeline Automates Testing, Build & Registry Publish

Pipeline Flow Components:

1. Push / PR (Trigger Event)

2. API Tests (Pytest Verification)

3. Docker Build (Python 3.11-slim)

4. Smoke Test (/api/health Endpoint)

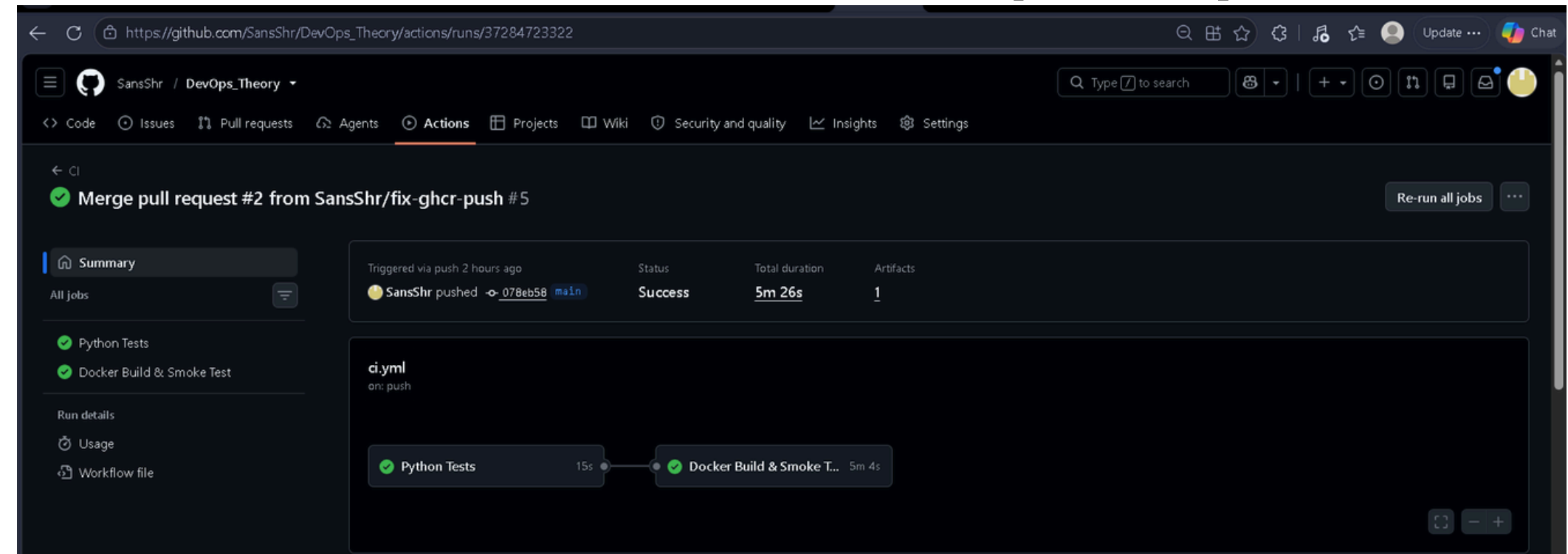
5. GHCR Publish (Container Registry)

6. K8s Deploy (Rolling Update)

CI/CD Pipeline Stages:

- Trigger: Automated on Code Push or Pull Request.
- Automated Testing: Pytest runs full API unit test suite.
- Smoke Test: Container verified against /api/health.
- GHCR Publish: Images pushed to GitHub Container Registry.

Successful GitHub Actions CI/CD Pipeline



Published CAMATIS Backend Image on GHCR

deployment/camatis-backend -n camatis

Successful Rollback and Deployment History

Prometheus Metrics & Provisioned Grafana Dashboards Track System Health

Editable Rollout Management Commands:

API Uptime

PromQL Query:
up{job="camatis-backend"}

Request Rate

PromQL Query:
rate(http_requests_total[5m])

P95 Latency

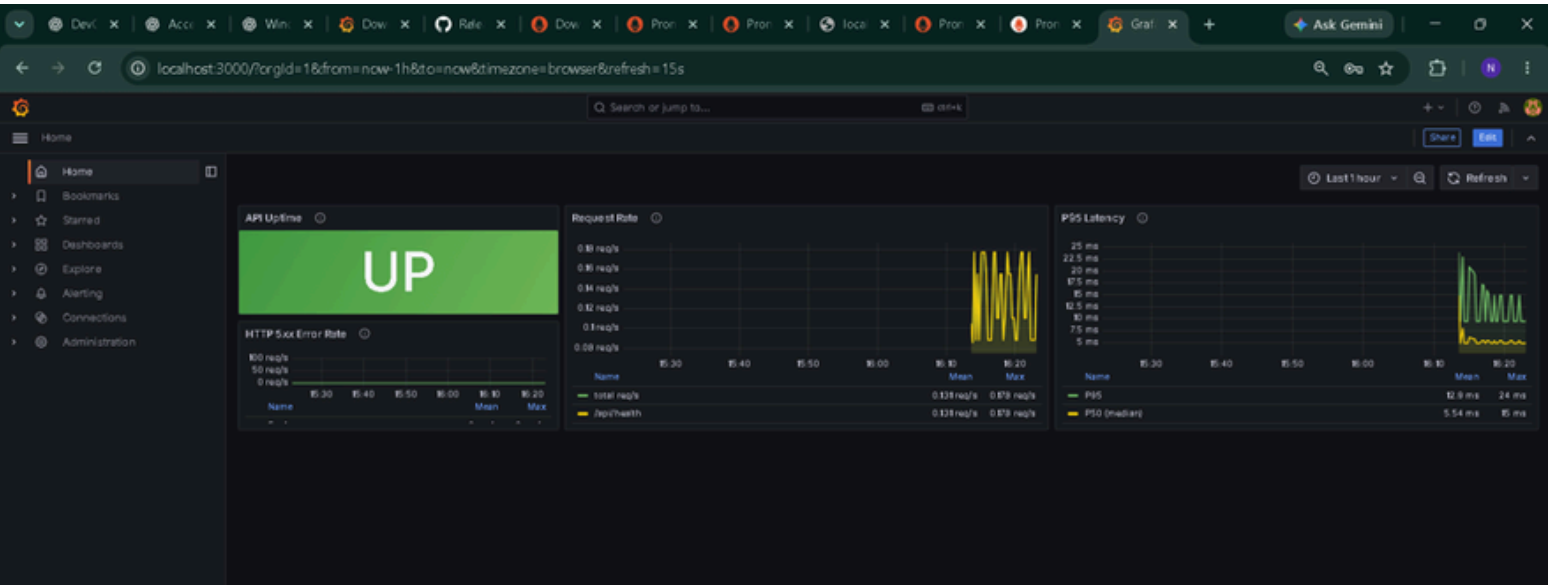
PPromQL Query:
histogram_quantile(0.95, ...)

5xx Errors

PromQL Query:
sum(rate(http_5xx_total[5m]))

Observability Architecture:

- Metrics Endpoint: Exposed at /metrics and scraped every 15s.
- Grafana Integration: Dashboards auto-provisioned from repo config.
- Log Inspection: Streamed directly via kubectl logs -n camatis.



Grafana Observability Dashboard

Prometheus Target Health: Backend & Prometheus UP

A screenshot of the Prometheus Targets page. The page shows a table of targets with columns for Endpoint, State, Labels, Last Scrape, Scrape Duration, and Error. Two targets are listed: 'camatis-backend (1/1 up)' and 'prometheus (1/1 up)'. Both targets are in a 'UP' state. The page includes a search bar and a filter button.

Lazy-Loading ML Pipelines & Automated Tooling

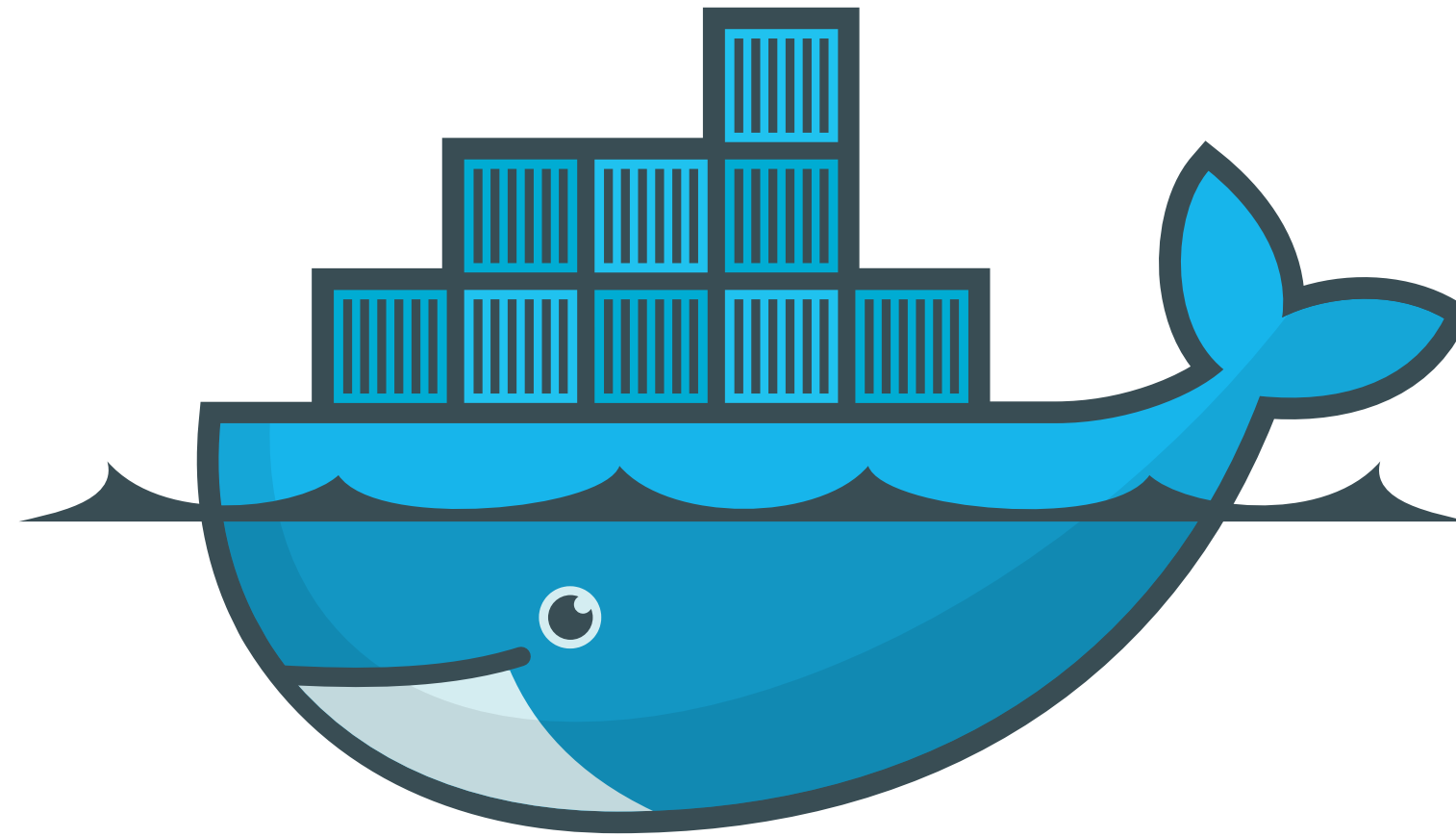
Optimized Performance

Challenge: ML Pipeline Startup Overhead
Importing heavy ML dependencies on API boot caused slow initial container startup and health probe timeouts.

Solution: Lazy-Loading Implementation
Deferred ML model initialization until first /api/optimize call, keeping boot ultra-fast and lightweight.

- Operational Insights & Best Practices**
1. Docker ensures machine-agnostic execution across environments.
 2. K8s rolling updates provide risk-free deployments.
 3. Ansible host automation eliminates manual setup mistakes.

```
Handling connection for 3000
PS C:\Users\Sanskriti\OneDrive\Desktop\CAMATIS Devops> kubectl logs -n camatis -l app=camatis-backend --tail=100 --prefix
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:34154 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:34162 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:53790 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:36726 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:39504 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:39510 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:33828 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.12:37048 - "GET /metrics HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:39152 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:50630 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:50636 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.12:41114 - "GET /metrics HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:35214 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.12:51618 - "GET /metrics HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:44300 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:55396 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:55410 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.12:47624 - "GET /metrics HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:58574 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:48310 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:33822 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:33830 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:53766 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:51076 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:39736 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:39742 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.12:49004 - "GET /metrics HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:32900 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.12:43218 - "GET /metrics HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:59896 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:44258 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:44274 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:46794 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.12:59344 - "GET /metrics HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:42224 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:34816 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:34824 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.12:38272 - "GET /metrics HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:35868 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:50410 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:53862 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:53874 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:59944 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.12:54206 - "GET /metrics HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:36248 - "GET /api/health HTTP/1.1" 200 OK
[pod/camatis-backend-56584c9948-qzqqk/camatis-backend] INFO: 10.244.0.1:50006 - "GET /api/health HTTP/1.1" 200 OK
```



Thank You